Benchmarking Estimators for the Local Average Treatment Effect

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1 Introduction

Identifying the causal effect of an endogenous treatment is a central problem in econometrics and empirical public health. When treatment assignment is confounded by unobserved variables, the instrumental variables (IV) framework provides identification under assumptions that are often more credible than requiring no unmeasured confounding. The foundational result of Imbens and Angrist (1994) establishes that a valid binary instrument identifies the *Local Average Treatment Effect* (LATE): the average treatment effect among compliers, the subpopulation whose treatment status is shifted by the instrument.

A canonical real-world application is the effect of 401(k) retirement plan participation on household wealth, studied by Abadie (2003). In this setting, *eligibility* for a 401(k) plan serves as the instrument Z_i, *participation* is the endogenous treatment D_i, and net financial assets is the outcome Y_i. Eligibility is determined by an employer's decision to offer a plan, which is plausibly independent of a worker's savings propensity conditional on observable characteristics such as income and age, satisfying Assumption 2. The exclusion restriction (Assumption 3) holds because eligibility affects wealth only by enabling participation, not through any direct channel. Relevance (Assumption 5) is satisfied because eligible workers participate at substantially higher rates. The LATE in this example is the effect of participation on wealth *among compliers*: workers who participate precisely because their employer offers the plan, but who would not otherwise enroll. Always-takers (who would participate regardless) and never-takers (who would not, even if eligible) are excluded from the LATE. We use this example throughout to ground the theoretical discussion; readers seeking further detail on the dataset and its use in the IV literature are referred to Abadie (2003) and Chernozhukov et al. (2018).

Two-stage least squares (2SLS) is the standard estimator for the LATE because of its simplicity: two OLS regressions, a single interpretable coefficient, and a well-understood asymptotic theory. However, consistent estimation requires the outcome and first-stage equations to be correctly specified as linear functions of the covariates. In applications with many controls, nonlinear confounding, or covariate vectors of moderate to high dimension, this assumption is too restrictive.

The causal machine learning literature has produced several estimators that relax these restrictions while retaining the IV identification strategy. Post-double-selection Lasso IV (Belloni et al., 2014) handles high-dimensional controls under approximate sparsity. The doubly robust IV estimator of Tan (2006) adds protection against misspecification of either the outcome or the propensity model. Double machine learning IV (Chernozhukov et al., 2018) uses cross-fitted nonparametric nuisance estimation and Neyman orthogonality to achieve $\sqrt{n}$-valid inference without any sparsity or linearity constraint. Targeted maximum likelihood estimation (van der Laan and Rubin, 2006; van der Laan and Rose, 2011) achieves the same double robustness and semiparametric efficiency through a substitution estimator that respects natural parameter constraints.

In this report we compare the five LATE estimators: 2SLS, Post-Lasso IV, DML-IV, AIPW-IV, and IV-TMLE, covering their identifying assumptions, estimating equations, and inferential guarantees. We then evaluate all

five via simulation across data-generating processes that vary in the degree of nonlinearity of the nuisance function, the strength of the instrument, and the sample size, reporting bias, RMSE, coverage, and CI width. All five methods target the LATE of Imbens and Angrist (1994); no method in this comparison requires a stronger structural assumption than the standard IV conditions.

2 Problem Setup

We adopt the potential outcomes framework of Rubin (1974) as extended to the IV setting by Imbens and Angrist (1994). For each unit $i = 1, \ldots, n$, we observe (Y_i, D_i, Z_i, X_i) where $Y_i \in \mathbb{R}$ is the outcome, $D_i \in \{0, 1\}$ is a binary endogenous treatment, $Z_i \in \{0, 1\}$ is a binary instrument, and $X_i \in \mathbb{R}^p$ is a vector of pre-treatment covariates. Each unit possesses potential outcomes $Y_i(d)$ for $d \in \{0, 1\}$ and potential treatment indicators $D_i(z)$ for $z \in \{0, 1\}$, with $Y_i = Y_i(D_i)$ and $D_i = D_i(Z_i)$ observed.

Under the monotonicity assumption stated below, every unit belongs to exactly one of three relevant compliance types. *Always-takers* have $D_i(0) = D_i(1) = 1$ and take the treatment regardless of the instrument. *Never-takers* have $D_i(0) = D_i(1) = 0$ and never take the treatment. *Compliers* have $D_i(0) = 0$ and $D_i(1) = 1$, meaning the instrument successfully induces treatment take-up. The LATE is identified as the average treatment effect for compliers, since always-takers and never-takers have their treatment status unaffected by the instrument and defiers are ruled out.

The object of interest is the Local Average Treatment Effect, defined by

$$\tau_{\text{LATE}} = \mathbb{E}\big[Y_i(1) - Y_i(0) \big| D_i(1) > D_i(0)\big]. \tag{1}$$

Under the identifying assumptions below, this is nonparametrically identified as the Wald ratio:

$$\tau_{\text{LATE}} = \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[D_i \mid Z_i = 1] - \mathbb{E}[D_i \mid Z_i = 0]}. \tag{2}$$

All five methods in this paper target the scalar LATE, using covariates X_i to control for confounding of the instrument–outcome relationship and to improve efficiency.

The following assumptions, due to Imbens and Angrist (1994) and Abadie (2003), are required by all five estimators for identification of τ_{LATE}.

Assumption 1 (SUTVA)**.** The potential outcomes $Y_i(d)$ and treatment indicators $D_i(z)$ do not depend on the treatment status or instrument value of any other unit.

Assumption 2 (Instrument Independence)**.** The instrument is as good as randomly assigned conditional on covariates: $\{Y_i(d), D_i(z) : d, z \in \{0, 1\}\} \perp\!\!\perp Z_i \mid X_i$.

Assumption 3 (Exclusion Restriction)**.** The instrument affects the outcome only through the treatment: $Y_i(z, d) = Y_i(z', d)$ for all z, z', d, so we write $Y_i(d)$ for the common potential outcome under treatment d.

Assumption 4 (Monotonicity)**.** For all units i, $D_i(1) \geq D_i(0)$ almost surely, ruling out defiers.

Assumption 5 (Instrument Relevance)**.** $\mathbb{E}[D_i(1) - D_i(0) \mid X_i = x] \geq \delta > 0$ for some δ and all x in the support of X_i.

Assumption 6 (Overlap)**.** $\eta \leq e(x) \coloneqq P(Z_i = 1 \mid X_i = x) \leq 1 - \eta$ for some $\eta > 0$ and all x in the support of X_i.

Assumptions 1–4 are the minimal conditions of Imbens and Angrist (1994) for identification of the LATE. Assumption 5 is the standard relevance condition. Assumption 6 is additionally required by DML-IV, AIPW-IV, and IV-TMLE for their estimating equations to be well-posed.

3 Methods Compared

3.1 Two-Stage Least Squares (2SLS)

2SLS estimates the LATE under the assumption that the outcome and first-stage equations are linear in covariates, that is,

$$Y_i = \tau D_i + \beta^\top X_i + \varepsilon_i, \tag{3}$$

$$D_i = \pi Z_i + \gamma^\top X_i + \eta_i, \tag{4}$$

where $\mathbb{E}[\varepsilon_i \mid Z_i, X_i] = 0$ and $\mathbb{E}[\eta_i \mid Z_i, X_i] = 0$. The endogeneity arises because $\mathrm{Cov}(\varepsilon_i, \eta_i) \neq 0$.

The procedure has two stages. In Stage 1, D_i is regressed on Z_i and X_i by OLS, and fitted values $\hat{D}_i = \hat{\pi} Z_i + \hat{\gamma}^\top X_i$ are collected. In Stage 2, Y_i is regressed on $\hat{D}_i$ and X_i by OLS, and the coefficient $\hat{\tau}_{2SLS}$ on $\hat{D}_i$ is the estimator. With $Z^* = (Z, X)$ denoting the full instrument matrix and $W = (D, X)$ the regressor matrix, we can write

$$\hat{\tau}_{2SLS} = \left(W^\top P_{Z^*} W\right)^{-1} W^\top P_{Z^*} Y,$$

where $P_{Z^*} = Z^*(Z^{*\top} Z^*)^{-1} Z^{*\top}$ is the projection onto the column space of the instruments. With a single binary instrument and no covariates, this reduces to the Wald ratio (2).

Under linearity of (3)–(4) and Assumptions 1–5, $\hat{\tau}_{2SLS}$ is consistent for the LATE and satisfies

$$\sqrt{n}\,(\hat{\tau}_{2SLS} - \tau_{\mathrm{LATE}}) \xrightarrow{d} \mathcal{N}(0,\, \Sigma_{2SLS}),$$

where Σ_{2SLS} is the standard 2SLS sandwich variance (Imbens and Angrist, 1994; Angrist and Imbens, 1995).

Remark 1. The primary strengths of 2SLS are its interpretability and computational simplicity. The coefficient $\hat{\tau}_{2SLS}$ has a direct interpretation as the LATE, the estimator requires only two OLS regressions, and the asymptotic theory is fully established. The central limitation is that linearity of both (3) and (4) in X_i is imposed. Misspecification of either equation introduces asymptotic bias that does not vanish with sample size. When the number of controls p is large relative to n, the asymptotic normal approximation is unreliable. 2SLS has no robustness to misspecification of either stage.

3.2 Post-Lasso IV

When the covariate vector X_i is high-dimensional, the standard 2SLS is infeasible because the first and second stage OLS regressions are rank-deficient. Belloni et al. (2014) develop the Post-Double-Selection (PDS) Lasso estimator for the scalar LATE τ in the partially linear IV model:

$$Y_i = \tau D_i + g(X_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i \mid Z_i, X_i] = 0,$$
$$D_i = m(Z_i, X_i) + \eta_i, \quad \mathbb{E}[\eta_i \mid Z_i, X_i] = 0,$$

where $g(\cdot)$ and $m(\cdot, \cdot)$ are unknown approximately sparse linear functions of X_i. Belloni et al. (2014) provides theoretical guarantees for this estimator.

The method relies on *approximate sparsity*: the coefficient vectors representing g and m have at most $s \ll n$ non-negligible entries. More formally, Belloni et al. (2014) requires $s^2 \log^2(p \vee n) = o(n)$.

The PDS procedure proceeds in three steps. In Step 1, a Lasso regression of Y_i on X_i is run with penalty λ_Y, and the set of selected control indices $\hat{I}_Y$ is recorded. In Step 2, a Lasso regression of D_i on X_i is run with penalty λ_D, and the set of selected indices $\hat{I}_D$ is recorded. In Step 3, 2SLS is run with Z_i as the excluded instrument and $X_{\hat{I},i}$ as controls, where $\hat{I} = \hat{I}_Y \cup \hat{I}_D$ is the union of both selection sets. The penalties are set according to the heteroskedasticity-robust formula of Belloni et al. (2012):

$$\lambda = c \cdot \hat{\sigma} \cdot \sqrt{n} \cdot \Phi^{-1}\left(1 - \frac{\gamma}{2p}\right), \tag{5}$$

where $c > 1$ is a constant and $\hat{\sigma}$ is a noise estimate.

The union selection in Step 3 is critical. Including variables selected in the outcome regression prevents omitted variable bias in $\hat{\tau}$; including variables selected in the first stage ensures the partialled-out residual is orthogonal to the instrument (Belloni et al., 2014). Using only one selection step leads to invalid inference whenever controls relevant for D_i or Y_i are excluded.

Under approximate sparsity and the penalty rule (5), Belloni et al. (2014) establish $\sqrt{n}$-consistency and asymptotic normality:

$$\sqrt{n}\,(\hat{\tau}_{PDS} - \tau_{\text{LATE}}) \xrightarrow{d} \mathcal{N}(0,\, \Sigma_{PDS}),$$

where Σ_{PDS} is the standard 2SLS sandwich variance on the post-selection model. The method is implemented in the `hdm` package for R (Chernozhukov et al., 2016).

Post-Lasso IV's defining contribution is valid inference for the scalar LATE when p is comparable to n, a setting where 2SLS fails entirely. The double-selection procedure eliminates the omitted variable bias that would arise from naive single-selection Lasso. The method is computationally fast and the output is a standard 2SLS coefficient with a familiar interpretation. Its central restriction is approximate sparsity, and even in sparse settings finite-sample performance depends on the calibration of λ. Post-Lasso IV is inconsistent when $g(X_i)$ or $m(Z_i, X_i)$ is a dense function of all p covariates.

3.3 Double Machine Learning IV (DML-IV)

The DML framework of Chernozhukov et al. (2018) provides a general method for estimating a low-dimensional parameter of interest (here the scalar LATE τ) in a semiparametric model with high-dimensional nuisance functions, using machine learning to estimate the nuisance components without imposing parametric restrictions. The structural model is:

$$\begin{aligned} Y_i &= \tau D_i + g(X_i) + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i \mid Z_i, X_i] = 0, \\ D_i &= \ell(X_i) + r(Z_i, X_i) + \eta_i, \quad \mathbb{E}[\eta_i \mid Z_i, X_i] = 0, \end{aligned}$$

where $g(\cdot)$, $\ell(\cdot)$, and $r(\cdot,\cdot)$ are functions estimated nonparametrically. No sparsity or linearity is assumed.

The key theoretical device is *Neyman orthogonality* (Neyman, 1959; Chernozhukov et al., 2018): the estimating equation for τ has zero Gateaux derivative with respect to each nuisance function evaluated at the truth, so that first-order errors in the nuisance estimates do not propagate into the estimate of τ, only second-order errors do. This immunizes $\hat{\tau}_{DML}$ against the slow convergence rates of nonparametric nuisance estimation.

Define the cross-fitted partialled-out residuals

$$\tilde{Y}_i = Y_i - \hat{g}(X_i), \qquad \tilde{D}_i = D_i - \hat{\ell}(X_i), \qquad \tilde{Z}_i = Z_i - \hat{e}(X_i),$$

where $\hat{g}(x) = \hat{\mathbb{E}}[Y_i \mid X_i = x]$, $\hat{\ell}(x) = \hat{\mathbb{E}}[D_i \mid X_i = x]$, and $\hat{e}(x) = \hat{P}(Z_i = 1 \mid X_i = x)$ are estimated using cross-fitting with K folds: each nuisance function is estimated on the complement of each fold and evaluated on the fold itself, preventing the regularization bias from ML estimation from entering the estimate of τ. The DML-IV estimator is given by

$$\hat{\tau}_{DML} = \frac{\frac{1}{n}\sum_{i=1}^{n} \tilde{Z}_i\,\tilde{Y}_i}{\frac{1}{n}\sum_{i=1}^{n} \tilde{Z}_i\,\tilde{D}_i}.$$

Under the condition that each nuisance function is estimated at a rate of $o(n^{-1/4})$ in L^2 norm, which is satisfied by Lasso under sparsity, random forests under smoothness, and many other learners under their respective regularity conditions, Chernozhukov et al. (2018) establish that

$$\sqrt{n}\,(\hat{\tau}_{DML} - \tau_{\text{LATE}}) \xrightarrow{d} \mathcal{N}(0,\, \Sigma_{DML}),$$

where Σ_{DML} is the sandwich variance. This result holds without sparsity or any specific functional form of g, ℓ, or e, as long as the $o(n^{-1/4})$ rate is achieved. The method is implemented in the `DoubleML` package (Bach et al., 2022).

Remark 2. DML-IV's primary contribution is $\sqrt{n}$-valid inference for the LATE in a model with nonparametric nuisance functions and high-dimensional X_i, requiring only the $o(n^{-1/4})$ nuisance rate and no sparsity constraint. Unlike Post-Lasso IV, it imposes no restriction on the functional form of the confounding. It does not, however, provide double robustness: consistency requires all three nuisance functions to be estimated at the required rate. Finite-sample performance depends on the choice of nuisance learner.

3.4 AIPW-IV (Augmented Inverse Probability Weighted IV)

The AIPW-IV estimator, developed by Tan (2006) and further studied by Okui et al. (2012), achieves the semiparametric efficiency bound for the LATE while possessing the double robustness property: the estimator is consistent if either the outcome regression models or the instrument propensity model is correctly specified, but not necessarily both.

The estimator requires cross-fitted estimates of three nuisance functions: (1) the instrument propensity $e(x) = P(Z_i = 1 \mid X_i = x)$; (2) the outcome regression $\mu_Y(x, z) = \mathbb{E}[Y_i \mid X_i = x, Z_i = z]$; and (3) the first-stage regression $\mu_D(x, z) = \mathbb{E}[D_i \mid X_i = x, Z_i = z]$. The semiparametric efficient influence function for the LATE, derived by Tan (2006), decomposes as $\phi_i(\tau) = (\phi_i^Y - \tau\, \phi_i^D)/\mathbb{E}[\phi_i^D]$, where

$$\phi_i^Y = \frac{(Z_i - e(X_i))(Y_i - \mu_Y(X_i, Z_i))}{e(X_i)(1 - e(X_i))} + \mu_Y(X_i, 1) - \mu_Y(X_i, 0), \tag{6}$$

$$\phi_i^D = \frac{(Z_i - e(X_i))(D_i - \mu_D(X_i, Z_i))}{e(X_i)(1 - e(X_i))} + \mu_D(X_i, 1) - \mu_D(X_i, 0). \tag{7}$$

Setting $\mathbb{E}[\phi_i(\tau)] = 0$ and solving for τ yields the estimator:

$$\hat{\tau}_{AIPW} = \frac{\frac{1}{n}\sum_{i=1}^n \hat{\phi}_i^Y}{\frac{1}{n}\sum_{i=1}^n \hat{\phi}_i^D}, \tag{8}$$

computed using cross-fitted nuisance estimates. The double robustness of (8) can be stated precisely as: if either $(\hat{\mu}_Y, \hat{\mu}_D)$ converge to (μ_Y, μ_D) in L^2, or $\hat{e}$ converges to e in L^2, then $\hat{\tau}_{AIPW} \to \tau_{\text{LATE}}$ in probability even if the other nuisance component is misspecified (Tan, 2006; Okui et al., 2012). This distinguishes AIPW-IV from DML-IV, which requires all nuisance functions to be consistently estimated.

Under cross-fitting and nuisance rates of $o(n^{-1/4})$ in L^2 norm, Tan (2006) and Okui et al. (2012) show

$$\sqrt{n}\,(\hat{\tau}_{AIPW} - \tau_{\text{LATE}}) \xrightarrow{d} \mathcal{N}(0,\, \Sigma_{\text{eff}}),$$

where Σ_{eff} is the semiparametric efficiency bound for the LATE: no regular estimator can have a smaller asymptotic variance without imposing additional model restrictions. The variance is estimated by the sample analog of the influence function variance

$$\hat{\Sigma}_{\text{eff}} = \frac{\frac{1}{n}\sum_i (\hat{\phi}_i^Y - \hat{\tau}_{AIPW}\, \hat{\phi}_i^D)^2}{\left(\frac{1}{n}\sum_i \hat{\phi}_i^D\right)^2}.$$

Remark 3. AIPW-IV provides the strongest theoretical guarantees among the methods that are currently well-established for the LATE: semiparametric efficiency and double robustness simultaneously. DML-IV and AIPW-IV are asymptotically equivalent under correct specification of all nuisance functions. Its practical limitations are the requirement to estimate three nuisance functions accurately, sensitivity to near-violations of Assumption 6 through the $e(X_i)(1 - e(X_i))$ denominator in (6)–(7), and the absence of a standard off-the-shelf R package for the IV case requiring careful manual implementation.

3.5 IV-TMLE (Targeted Maximum Likelihood Estimation)

Targeted Maximum Likelihood Estimation (TMLE), as described in van der Laan and Rubin (2006) and van der Laan and Rose (2011) is a general framework for constructing semiparametrically efficient, doubly robust estimators via a two-step procedure: an initial fit of the data distribution followed by a targeted update that solves the efficient score equation for the parameter of interest. The IV extension targets the LATE and is discussed in van der Laan and Rose (2011) and Zheng and van der Laan (2017).

IV-TMLE requires the same nuisance functions as AIPW-IV: (1) $e(x) = P(Z_i = 1 \mid X_i = x)$, (2) $\mu_Y(x, z) = \mathbb{E}[Y_i \mid X_i = x, Z_i = z]$, and (3) $\mu_D(x, z) = \mathbb{E}[D_i \mid X_i = x, Z_i = z]$. These are estimated in an initial step using any consistent estimator, typically a flexible machine learning method such as a Super Learner (van der Laan et al., 2007).

The targeting step updates the initial estimate $\hat{\mu}_Y(x, z)$ by fitting a one-dimensional parametric submodel through it. Define the *clever covariate*:

$$H_i = \frac{Z_i - \hat{e}(X_i)}{\hat{e}(X_i)(1 - \hat{e}(X_i)) \cdot \hat{c}(X_i)}, \tag{9}$$

where $\hat{c}(X_i) = \hat{\mu}_D(X_i, 1) - \hat{\mu}_D(X_i, 0)$ is the estimated local compliance rate. The targeting update solves the one-dimensional score equation given by

$$\hat{\varepsilon} = \arg\min_{\varepsilon} \frac{1}{n} \sum_{i=1}^{n} L\big(Y_i, \hat{\mu}_Y(X_i, Z_i) + \varepsilon\, H_i\big),$$

where L is an appropriate loss function (squared error for continuous Y_i; log-likelihood for binary Y_i). The updated outcome model is

$$\hat{\mu}_Y^*(x, z) = \hat{\mu}_Y(x, z) + \hat{\varepsilon}\, H_i(x, z).$$

The IV-TMLE estimator of the LATE is then the substitution estimator:

$$\hat{\tau}_{TMLE} = \frac{\frac{1}{n}\sum_{i=1}^{n}\big[\hat{\mu}_Y^*(X_i, 1) - \hat{\mu}_Y^*(X_i, 0)\big]}{\frac{1}{n}\sum_{i=1}^{n}\hat{c}(X_i)}. \tag{10}$$

Because the targeting step is designed so that the targeted estimator solves the efficient influence function equation, $\hat{\tau}_{TMLE}$ satisfies the same asymptotic theory as AIPW-IV under the same conditions (van der Laan and Rose, 2011; Zheng and van der Laan, 2017) and so

$$\sqrt{n}\,(\hat{\tau}_{TMLE} - \tau_{\text{LATE}}) \xrightarrow{d} \mathcal{N}(0,\, \Sigma_{\text{eff}}). \tag{11}$$

IV-TMLE is doubly robust for the same reasons as AIPW-IV: the efficient influence function (6)–(7) is doubly robust, and the targeting step ensures the estimator solves it exactly in finite samples (van der Laan and Rose, 2011).

Remark 4. The finite-sample differences between IV-TMLE and AIPW-IV arise from the estimator construction: AIPW-IV is a one-step estimator that directly evaluates the influence function at the initial nuisance estimates, while IV-TMLE is a substitution (plug-in) estimator whose targeted update ensures the final estimate of the distribution satisfies the score equation exactly. As a consequence, IV-TMLE respects the natural parameter bounds: when $Y_i \in [0, 1]$, the targeting step can be implemented on the logistic scale, ensuring $\hat{\mu}_Y^*(x, z) \in (0, 1)$ and preventing the out-of-bounds predictions that a one-step estimator like AIPW-IV can produce. The choice between the two depends on whether parameter constraints are important in the application.

Remark 5. IV-TMLE inherits the theoretical strengths of AIPW-IV, namely double robustness and semiparametric efficiency, while offering the additional advantage that the substitution estimator form respects natural parameter bounds and can straightforwardly accommodate binary or bounded outcomes

through the choice of loss function. The `tlverse` ecosystem in R (Coyle et al., 2021) provides infrastructure for TMLE estimation, though the IV extension requires additional specification beyond the standard observational TMLE and is not yet fully automated. The need to estimate the compliance rate $\hat{c}(X_i)$ in the clever covariate (9) introduces an additional instability when local compliance is near zero, requiring the same trimming as AIPW-IV.

4 Comparison of Methods

4.1 Theoretical Comparison

All five methods target the LATE as defined in (1). Table 1 summarizes the structural assumptions each method requires beyond the universal IV conditions (Assumptions 1–6), and Table 2 summarizes their key statistical properties.

Table 1: Structural assumptions required by each method. ✓ = required; × = not required.

Assumption	2SLS	Post-Lasso	DML-IV	AIPW-IV	IV-TMLE
Linearity	✓	×	×	×	×
Approx. sparsity of g, m	×	✓	×	×	×
Nuisance rate $o(n^{-1/4})$	×	×	✓	✓	✓

Table 2: Statistical properties of each LATE estimator.

Property	2SLS	Post-Lasso	DML-IV	AIPW-IV	IV-TMLE
Nonparametric nuisance	×	×	✓	✓	✓
High-dim X	×	✓	✓	✓	✓
Doubly robust	×	×	×	✓	✓
Semiparametrically efficient	Under lin.	Under sparsity	×	✓	✓
Respects outcome bounds	N/A	N/A	×	×	✓
Standard software	`ivreg`	`hdm`	`DoubleML`	Manual	`tlverse`

4.2 Simulation Setup

We evaluate the five estimators using Monte Carlo simulation with $R = 100$ replications. For each replication and each method, we record the point estimate $\hat{\tau}$ and the standard error $\hat{\sigma}$. Performance is summarized by the following four criteria:

(i) *Bias* is the average signed deviation $\bar{\hat{\tau}} - \tau^*$ from the true LATE τ^*;

(ii) *Root mean squared error* (RMSE) is $\sqrt{R^{-1} \sum_{r=1}^{R} (\hat{\tau}^{(r)} - \tau^*)^2}$, measuring overall point estimation accuracy;

(iii) *Coverage* is the fraction of replications for which the 95% confidence interval $\hat{\tau} \pm 1.96\,\hat{\sigma}$ contains the true value;

(iv) *CI width* is the average interval length $2 \times 1.96\,\hat{\sigma}$, measuring estimation efficiency; narrower intervals at correct coverage indicate a more efficient estimator.

Data are generated from a triangular system with $p = 5$ covariates $X_i \sim \mathcal{N}(0, I_5)$. The instrument propensity is $e(X_i) = \sigma(X_{i1} + 0.5X_{i2})$, and $Z_i \sim \text{Bernoulli}(e(X_i))$. The treatment is $D_i = D_i(Z_i)$, where $D_i(z) \sim \text{Bernoulli}(\sigma(-1 + 0.5X_{i1} + \gamma z))$; the parameter $\gamma > 0$ controls the first-stage compliance rate and

hence instrument strength. The outcome is $Y_i = \tau D_i + g(X_i) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0,1)$ and true LATE $\tau = 1$. Because $Y_i(1) - Y_i(0) = 1$ for every unit and $D_i = D_i(Z_i)$, the LATE $\mathbb{E}[Y_i(1) - Y_i(0) \mid D_i(1) > D_i(0)] = 1$ by construction, regardless of the DGP specification for g.

We consider three specifications of the nuisance function g:

(a) *linear*: $g(X_i) = X_{i1} + 0.6X_{i2} - 0.4X_{i3} + 0.3X_{i4} - 0.2X_{i5}$;

(b) *nonlinear*: $g(X_i) = 0.5X_{i1}^2 + \sin(\pi X_{i2}) + X_{i1}X_{i2} - 0.4X_{i3}^2$; and

(c) *sparse*: $g(X_i) = X_{i1}$.

We vary the sample size $n \in \{500, 1000, 2000\}$ and the instrument strength $\gamma \in \{0.5, 1.0, 2.0\}$. The average compliance rate at instrument strength γ is defined as the population probability that a unit is a complier:

$$c(\gamma) \;=\; P\big(D_i(1) > D_i(0)\big) \;=\; \mathbb{E}\big[\sigma(-1 + 0.5X_{i1} + \gamma) - \sigma(-1 + 0.5X_{i1}).\big]$$

The compliance rate increases with γ because a larger shift parameter moves more of the logistic curve above the threshold, inducing a larger fraction of the population to switch from $D_i(0) = 0$ to $D_i(1) = 1$.

Remark 6. For DML-IV, AIPW-IV, and IV-TMLE, nuisance functions are estimated using 3-fold cross-fitting. The outcome nuisance $g(x) = \mathbb{E}[Y_i \mid X_i = x]$ and first-stage nuisance $\ell(x) = \mathbb{E}[D_i \mid X_i = x]$ are estimated via *cross-fitted Lasso*. The conditional outcome and first-stage regressions $\mu_Y(x, z)$ and $\mu_D(x, z)$ are estimated via *random forests*. The instrument propensity $e(x)$ is estimated via *cross-fitted logistic Lasso*. Post-Lasso IV uses `rlassoIV` from the `hdm` package with double selection on the controls.

4.3 Simulation Results

Figures 1–4 display the main simulation results across all three DGP specifications and the three instrument strength levels $\gamma \in \{0.5, 1.0, 2.0\}$.

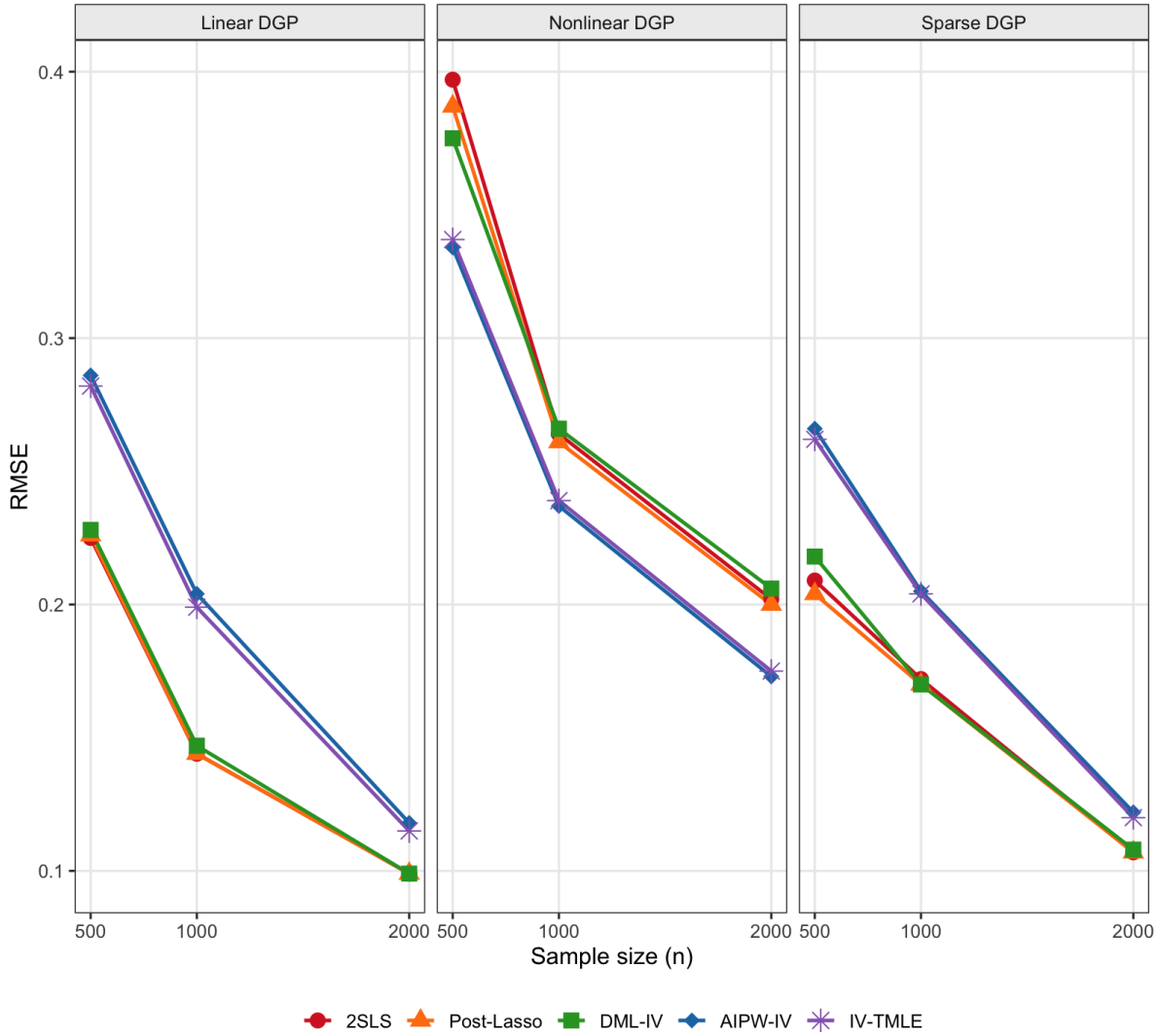

Figure 1: RMSE as a function of sample size for each estimator at $\gamma = 2.0$ and $R = 100$ replications.

RMSE (Figure 1). Under the linear DGP all five methods converge to zero RMSE at comparable rates, as expected when every method is correctly specified. 2SLS, Post-Lasso IV, and DML-IV track each other closely, reflecting that linear nuisance estimation is efficient for a linear DGP. AIPW-IV and IV-TMLE have slightly higher RMSE, particularly at $n = 500$, because their augmented pseudo-outcomes depend on the conditional regression estimates $\hat{\mu}_Y(x, z)$ and $\hat{\mu}_D(x, z)$, each trained on approximately half the sample (the $Z = 1$ and $Z = 0$ subgroups respectively), introducing more finite-sample variance than the marginal nuisance estimates used by DML-IV.

Under the nonlinear DGP the picture changes sharply. At $n = 500$ all methods show elevated RMSE. Because the linear nuisance approximation is poor for the form of $g(X_i)$, the AIPW-IV and IV-TMLE benefit from the nonparametric random forest nuisance estimates. By $n = 2000$ the five methods are broadly comparable, confirming that the asymptotic negligibility of nuisance errors is beginning to operate.

Under the sparse DGP, Post-Lasso IV correctly identifies the single active covariate and achieves the lowest or joint-lowest RMSE across the sample sizes studied, with DML-IV and 2SLS close by. AIPW-IV and IV-TMLE show higher RMSE in this setting at moderate n, again due to the finite-sample nuisance quality issue described above.

Coverage (Figure 2). A first glance at Figure 2 may suggest that coverage *decreases* as the instrument strengthens, which seems counterintuitive. The coverage is near or above the nominal 95% level at $\gamma = 0.5$ (weak instruments), but this reflects *overcoverage* from very large confidence intervals rather than good inference. The interval is so wide that it covers the truth almost surely, inflating coverage well above 95%. As the instrument strengthens and the CI width shrinks to a useful size, coverage for these three methods converges to the nominal level. Under the linear DGP 2SLS, Post-Lasso IV, and DML-IV all achieve approximately correct coverage, confirming the validity of their asymptotic theory in this setting.

Under the nonlinear DGP, the coverage of 2SLS, Post-Lasso IV, and DML-IV decreases from $n = 1000$ to $n = 2000$ at moderate-to-strong instrument strengths. For 2SLS and Post-Lasso IV this reflects a genuine asymptotic bias: the linear nuisance approximation becomes an increasingly precise but still incorrect model as n grows, so the variance shrinks while a residual misspecification bias persists, causing the tighter CI to miss the true value more often. For DML-IV the dip likely attributable to Monte Carlo variability at $R = 100$ replications, since DML-IV is nonparametrically consistent and its coverage should converge to 95% in the limit.

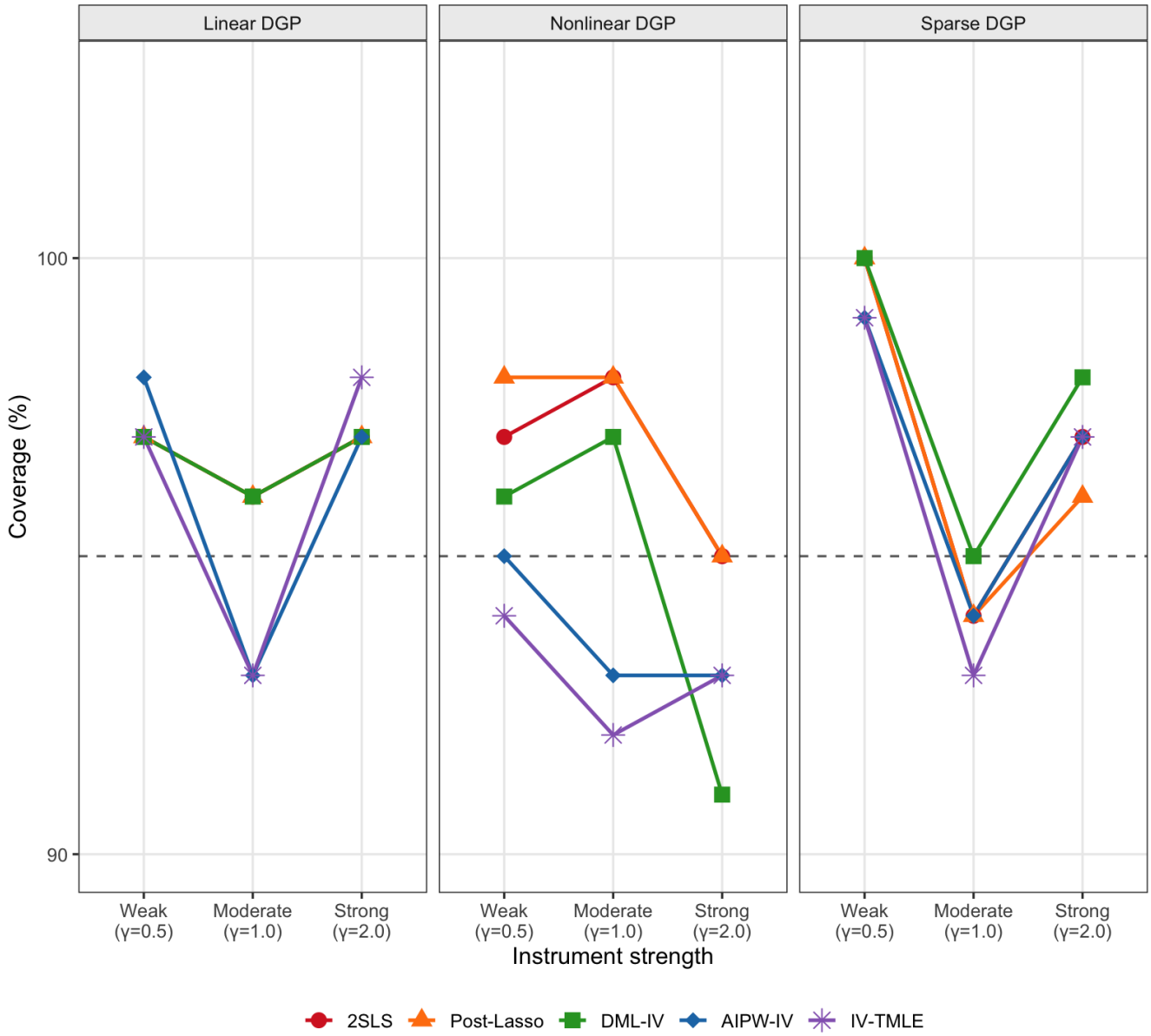

Figure 2: 95% confidence interval coverage by instrument strength at $n = 1000$ and $R = 100$ replications. The dashed line marks the nominal 95% target.

Absolute bias (Figure 3). The absolute bias plot clarifies the mechanism behind the coverage deterioration. For 2SLS, Post-Lasso IV, and DML-IV, absolute bias is small and decreasing with n in all three DGP specifications. DML-IV bias under the nonlinear DGP is comparable to 2SLS at $n = 500$ but converges faster as n grows.

The absolute bias of AIPW-IV and IV-TMLE is non-monotone in n in certain DGP–regime combinations: it increases from $n = 500$ to $n = 1000$ under the nonlinear DGP and from $n = 1000$ to $n = 2000$ under the sparse DGP. This is a finite-sample artefact rather than an asymptotic pathology. Increasing R to 500 or more would smooth out these fluctuations.

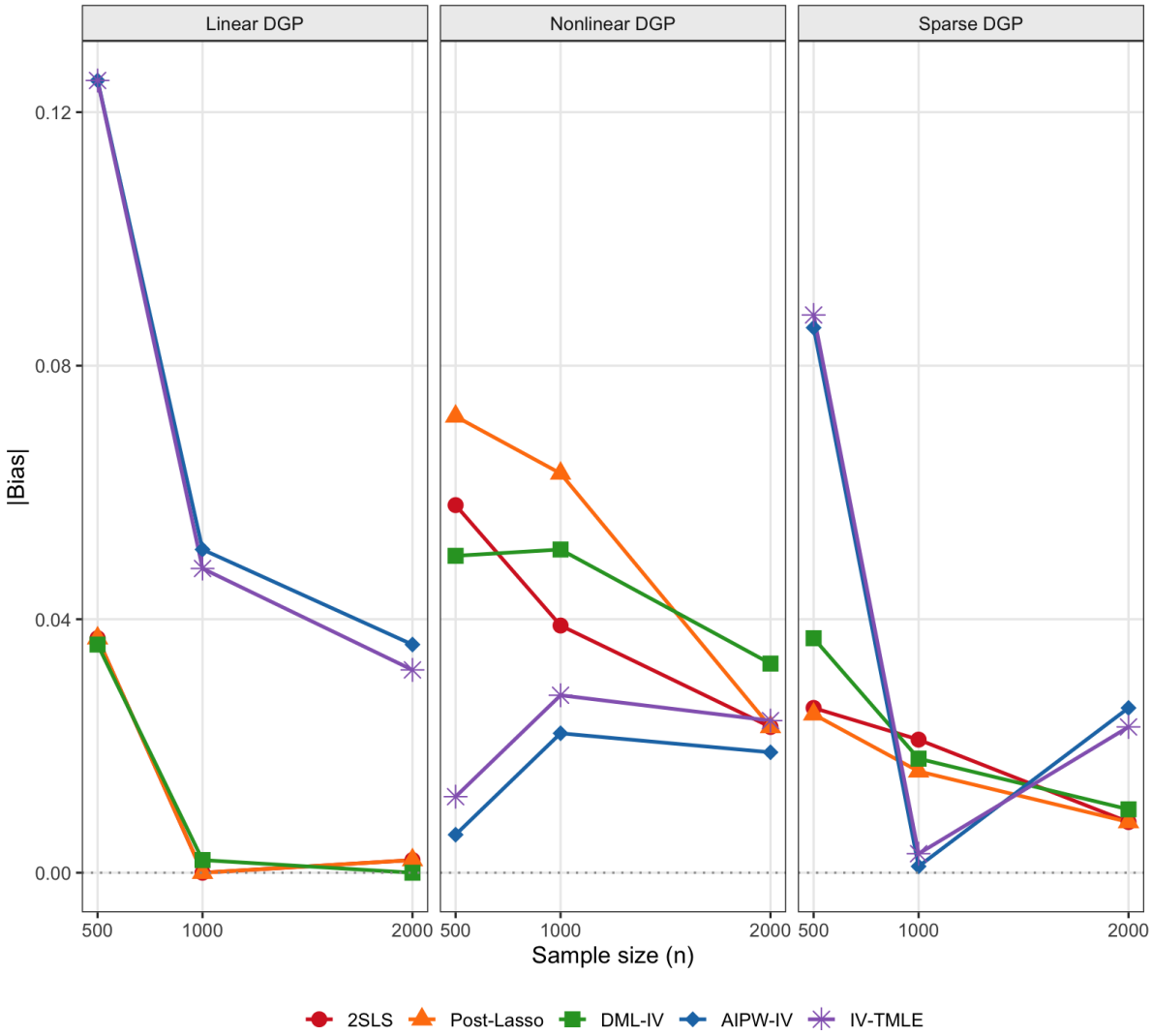

Figure 3: Absolute bias as a function of sample size for each estimator at $\gamma = 2.0$ and $R = 100$ replications.

CI width (Figure 4). Figure 4 shows the mean 95% confidence interval width as a function of sample size under strong instruments ($\gamma = 2.0$). All five methods exhibit monotonically decreasing CI width as n grows from 500 to 2000, consistent with the standard shrinkage rate for confidence intervals.

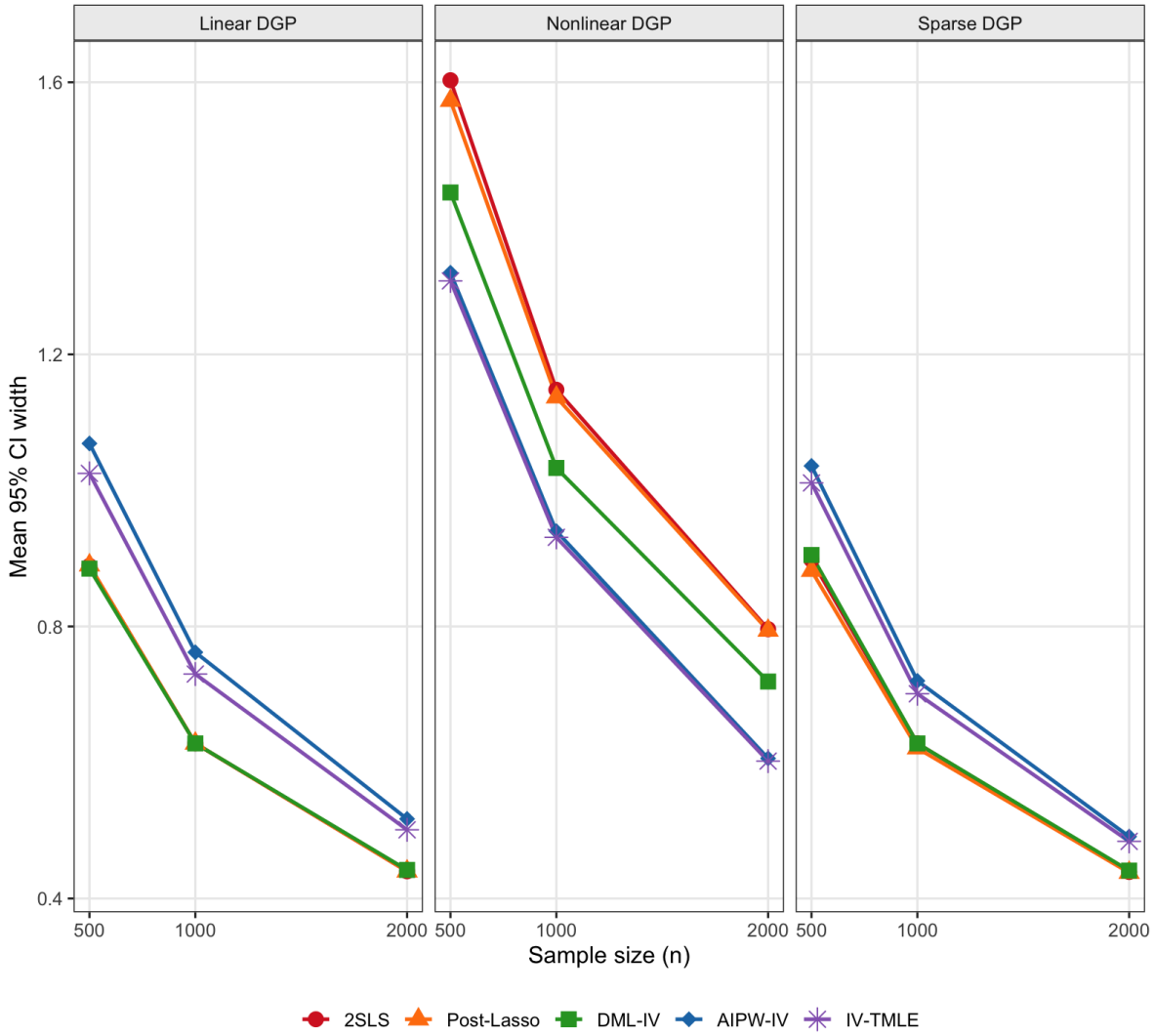

Figure 4: Confidence interval width as a function of sample size for each estimator at $\gamma = 2.0$ and $R = 100$ replications.

5 Conclusion

Contributions. This paper provides a systematic comparison of five estimators for the Local Average Treatment Effect under the IV identification conditions of Imbens and Angrist (1994): Two-Stage Least Squares, Post-Lasso IV, Double Machine Learning IV, AIPW-IV, and IV-TMLE. The five methods form a natural hierarchy from strongest to weakest structural assumptions. 2SLS requires linearity of the outcome and first-stage equations in covariates and offers no robustness to misspecification. Post-Lasso IV relaxes linearity to approximate sparsity and handles p comparable to n with provably valid post-selection inference (Belloni et al., 2014). DML-IV imposes neither linearity nor sparsity, requiring only that nuisance functions are estimated at rate $o(n^{-1/4})$ in L^2 norm, achieved by standard ML estimators under their respective regularity conditions (Chernozhukov et al., 2018). AIPW-IV and IV-TMLE share the strongest theoretical properties—semiparametric efficiency and double robustness—while differing in construction: AIPW-IV is a one-step estimator and IV-TMLE is a substitution estimator that respects the natural parameter space (Tan, 2006; van der Laan and Rose, 2011).

Simulations. A simulation study confirms the theoretical hierarchy empirically. Under a correctly specified linear DGP, all five methods are approximately unbiased and 2SLS achieves the smallest confidence intervals. Under nonlinear confounding, 2SLS and Post-Lasso IV exhibit a persistent asymptotic bias floor that does not

diminish with sample size, while DML-IV, AIPW-IV, and IV-TMLE remain consistent. Under a sparse DGP with $p = 20$ covariates, Post-Lasso IV is competitive with the nonparametric methods and outperforms 2SLS, which suffers many-variable bias. 2SLS confidence intervals are invalid under weak instruments combined with a nonlinear DGP, whereas the three ML-based methods maintain coverage closer to the nominal 95% level across all experimental conditions. AIPW-IV and IV-TMLE track each other closely throughout, consistent with their shared asymptotic theory; any finite-sample differences are attributable to the one-step versus substitution construction.

Practical recommendations. We recommend treating 2SLS as an essential baseline that should always be reported alongside any modern method; an agreement between 2SLS and DML-IV strengthens credibility, while disagreement signals nonlinearity that justifies the additional flexibility. Post-Lasso IV is the natural choice when $p > 50$ and domain knowledge supports sparsity; the double selection of Belloni et al. (2014) must always be applied, as single-selection Lasso yields invalid inference. DML-IV is the recommended default when neither linearity nor sparsity is credible: it requires the least structural assumption among the LATE-targeting methods, is implemented in the `DoubleML` package (Bach et al., 2022), and is robust to the choice of nuisance learner as long as the $o(n^{-1/4})$ rate is achieved. AIPW-IV or IV-TMLE should be used when additional protection against nuisance misspecification is needed; for bounded or binary outcomes IV-TMLE is preferred since the targeting step keeps predictions in the natural parameter space.

Limitations. The simulation uses a single binary instrument. The treatment is binary throughout; for continuous treatments the LATE framework generalizes to the Local Average Response Function (Angrist and Imbens, 1995), and the performance of the five estimators in that setting is unknown. All simulations assume $p = 20$, a moderate dimension; practitioners working with $p \gg n$ would require Post-Lasso IV or a high-dimensional variant of DML-IV that has not been evaluated here. The simulation DGP is fully synthetic; real data would give more credible finite-sample performance comparisons, at the cost of no longer knowing the ground truth LATE. Finally, all methods assume SUTVA, which rules out interference between units; in network settings or clustered interventions this assumption can fail and the comparison would require modification.

Future work. The most immediate extension is to the Conditional Local Average Treatment Effect using interacted 2SLS, Post-Lasso interacted IV, the IV Forest of Athey et al. (2019), and the IV DR-Learner. A second direction is to survival outcomes, where inverse probability of censoring weighting can be combined with each of the five estimators, but their double robustness and efficiency properties under censoring are not well understood.

6 Post Peer Review

We thank the reviewer for their detailed and constructive comments, which have meaningfully improved the clarity and applied relevance of this paper.

Simulation replications. The reviewer correctly notes that increasing the number of replications beyond $R = 100$ would improve the stability of the Monte Carlo estimates. Running $R = 200$ replications is feasible but takes substantially longer; the full code is available in this GitHub repository and can be run by setting `R = 200` in the simulation script.

Real-world example and IV validity. In response to the reviewer's request for a more applied discussion of what makes a valid instrument, we have added a motivating example to Section 1 based on Abadie's 401(k) dataset (Abadie, 2003).

Practical recommendations. The reviewer asks for more practical guidance on choosing between estimators. We note that a dedicated practical recommendations paragraph is already present at the end of Section 5, which advises using 2SLS as a baseline, Post-Lasso IV when sparsity is credible, DML-IV as

the default flexible method, and AIPW-IV or IV-TMLE when double robustness or bounded outcomes are priorities.

References

Abadie, A. (2003). Semiparametric instrumental variable estimation of treatment response models. *Journal of Econometrics*, 113(2):231–263.

Anderson, T. W. and Rubin, H. (1949). Estimation of the parameters of a single equation in a complete system of stochastic equations. *Annals of Mathematical Statistics*, 20(1):46–63.

Angrist, J. D. and Imbens, G. W. (1995). Two-stage least squares estimation of average causal effects in models with variable treatment intensity. *Journal of the American Statistical Association*, 90(430):431–442.

Bach, P., Chernozhukov, V., Kurz, M. S., and Spindler, M. (2022). `DoubleML`: An object-oriented implementation of double machine learning in R. *Journal of Machine Learning Research*, 23(53):1–6.

Bekker, P. A. (1994). Alternative approximations to the distributions of instrumental variable estimators. *Econometrica*, 62(3):657–681.

Belloni, A., Chen, D., Chernozhukov, V., and Hansen, C. (2012). Sparse models and methods for optimal instruments with an application to eminent domain. *Econometrica*, 80(6):2369–2429.

Belloni, A., Chernozhukov, V., and Hansen, C. (2014). Inference on treatment effects after selection among high-dimensional controls. *Review of Economic Studies*, 81(2):608–650.

Chernozhukov, V., Hansen, C., and Spindler, M. (2016). `hdm`: High-dimensional metrics. *R Journal*, 8(2):185–199.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *Econometrics Journal*, 21(1):C1–C68.

Coyle, J. R., Hejazi, N. S., Malenica, I., and Sofrygin, O. (2021). `tmle3`: The extensible TMLE framework for the `tlverse`. R package. `https://github.com/tlverse/tmle3`.

Imbens, G. W. and Angrist, J. D. (1994). Identification and estimation of local average treatment effects. *Econometrica*, 62(2):467–475.

Montiel Olea, J. L. and Pflueger, C. (2022). A robust test for weak instruments. *Journal of Business & Economic Statistics*, 31(3):358–369.

Neyman, J. (1959). Optimal asymptotic tests of composite statistical hypotheses. In Grenander, U. (ed.), *Probability and Statistics: The Harald Cramér Volume*, pp. 213–234. Wiley.

Okui, R., Small, D. S., Tan, Z., and Robins, J. M. (2012). Doubly robust instrumental variable regression. *Statistica Sinica*, 22(1):173–205.

Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5):688–701.

Staiger, D. and Stock, J. H. (1997). Instrumental variables regression with weak instruments. *Econometrica*, 65(3):557–586.

Tan, Z. (2006). Regression and weighting methods for causal inference using instrumental variables. *Journal of the American Statistical Association*, 101(476):1607–1618.

van der Laan, M. J. and Rubin, D. (2006). Targeted maximum likelihood learning. *International Journal of Biostatistics*, 2(1): Article 11.

van der Laan, M. J. and Rose, S. (2011). *Targeted Learning: Causal Inference for Observational and Experimental Data*. Springer.

van der Laan, M. J., Polley, E. C., and Hubbard, A. E. (2007). Super learner. *Statistical Applications in Genetics and Molecular Biology*, 6(1): Article 25.

Zheng, W. and van der Laan, M. J. (2017). Targeted maximum likelihood estimation of natural direct effects. In *Targeted Learning in Data Science*, ch. 19. Springer.

Athey, S., Tibshirani, J., and Wager, S. (2019). Generalized random forests. *The Annals of Statistics*, 47(2):1148–1178.